

Digital Economics

The Commerce of Information in the 21st Century

First Edition

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To my father, who once told a younger and more foolish version of myself,
“You can’t survive without money.”

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Introduction

SURPRISINGLY, THERE ARE VERY FEW BOOKS that explore the intersection of economics and the current digital reality that impacts each of our lives. This is especially remarkable when one realizes that all of the members that make up the Group of Seven (G7)—the seven largest advanced economies in the world—rely on nontangible information goods to such an extent that digital information makes up 70% of each country's GDP.¹ Usually, when one tries to explore the question of digital information and its economic implications on the economy, one notices that the literature collects around three topics:

- The role of consumer information in the economy, which in the online world reduces to the role of network effects in the economy.
- The role of e-commerce in the economy, which amounts to digital or online marketing.
- The role of information and communication technology (ICT), which focuses on the technological side and emphasizes the role of computer science and the Internet of things (IOT) in the economy.

While these topics are very important, few books exist that bring them together and explore the overall impact digital information goods have on the economic situation we now find ourselves in the twenty-first century. This volume seeks to fill this absence, which is especially needed at the undergraduate level, by providing some of the most important readings in the field.

But what exactly is information, and what kind of impact does it have on the science of economics? For the purpose of this book, *information* is defined as anything that can be reduced to bits, digitally. The world now runs on the production, dissemination, and consumption of digital information. All of us constantly access all sorts of information, through all sorts of devices, from all sorts of providers. We read and interact with websites, we query databases, and we communicate with each other via social media. These sorts of activities permeate both our personal and professional lives. In order to successfully navigate this digital world, information consumers, information producers, and information policymakers need to understand what sorts of *information goods* are likely to be available and how much they are likely to cost.

¹ Luciano Floridi, *Information: A Very Short Introduction* (Oxford: Oxford University Press, 2010).

In their book, Shapiro and Varian remind us that while technology changes, economic laws do not.² Economics will always be the study of how limited resources are used and the social cooperation that motivates individuals to produce and distribute goods in markets. The basic tools of economics—demand, supply, marginal cost, opportunity cost, and marginal benefit will still be used to guide the financial and policy decisions of information producers. At the same time, however, information goods have unique properties that physical goods do not. For instance, while only one person at a time can drive a particular car or eat a particular hamburger, millions of people can simultaneously read the same book, listen to the same song, and use the same software. Economists call these properties *nonrivalrous* and *nonexcludable* (which is why copyright law is such an important topic these days). In addition, once information is produced the cost of copying it is negligible (what economists call *zero marginal costs*). These are issues that are part and parcel to living, working, purchasing, and being entertained in an information society; these are the issues addressed in this book.

The intersection of economics and the information society is complicated. Nonetheless, by examining the overall economic structure in which the commerce of information exists, exploring important issues regarding digitized business models, and specific current trends such as Bitcoin and blockchain techniques, this book will further clarify how digital commerce functions in the twenty-first century.

2 Carl Shapiro and Hal R. Varian, *Information Rules: A Strategic Guide to the Network Economy* (Boston, Harvard Business School Press, 1999).

UNIT I

The Information Economy

THIS UNIT FOCUSES ON THE NATURE of information in the digital economy, the unique properties of information goods, and the market structures of information producers. Information, of course, has always been important in economics. In the digital economy we find ourselves in today, information is now seen as a particular kind of good in the marketplace. Information goods are different from physical goods and have their own unique economic properties. For the purposes of this book, we use the definition of *information* as anything that has been digitized and commoditized (either as a good or service). Information goods, then, are simply the digital products we use every day, such as software, operating systems, digital books, or digitized movies, and the digital platforms we utilize to access them. Today, information producers have become the dominant firms in the economic marketplace (think of the enormous roles that Amazon, Facebook, and Google play in bringing you information products). In addition to e-commerce, nearly every level of business, education, entertainment, health care, finance, and banking rely on the exchange of information products and the networked communications technology that provides these services. This is why information goods have become so important to the information economy and why digital information has become the dominant driver of economics globally. Furthermore, there are many processes in the information market. These processes center around the occurrence (the creation or authoring), transmission (networking, distributing, access), processing and management (collecting, validating, indexing, and filtering), and usage (monitoring, modeling, analyzing) of information goods. These are central operations of every business and firm in the information economy. Without them, networked economics will not function, and firms spend bounteously to make sure these processes work seamlessly.

Unlike physical goods, information has particular properties. Information goods are both nonrivalrous and nonexclusive. While many people can enjoy a digitized song, movie, or book,

only one person at a time can enjoy a hamburger or drive a car. This points to the nonexclusive character of digital information. It is very hard to prevent someone from using or gaining access to digital information. On the other hand, when an information good becomes inaccessible or corrupted for some reason, it has the potential to affect a very large number of people. Information goods are considered nonrivalrous because more than one person can share the benefit of the consumption of the same information product.

Another significant fact that distinguishes information goods from physical goods is that nearly all the costs are on the production side of the information good, and there are almost no costs at the reproduction and distribution side of the information good. What economists call “sunk costs” all occur at the production of the information good. Information goods can be very expensive to produce. However, the marginal costs of digital duplication and distribution are nearly zero. This increases the overall profitability of information goods and creates an interesting phenomenon among competitors in the digital marketplace. The competition among many information producers has made information cost less and less, sometimes even becoming free. Economists call this phenomenon *zero marginal cost* because the lowest marginal cost of most information goods on the Internet is free.

This condition of such freely available information has set up interesting new market structures. The main market structures we see today are the dominant firm, product differentiation, and cost leadership. For good or bad, digital network economies tend toward monopolies, which is why Facebook, Amazon, and Google and a few others have become dominant firms in the information economy. If an information producer wants to compete against monopolies, then it must either differentiate its product by adding unique value to its information good or service or by achieving cost leadership through network effects and scale. To compete in the information market, it is important to understand the market structure an information good falls into. These are the important themes that will be explored in this unit.

UNIT II

Specific Issues Within the Information Economy

THIS UNIT FOCUSES ON THE IMPORTANT role of product versioning and the kinds of business models most often found in the information marketplace. *Product versioning* is simply the way information producers differentiate their information product for various markets. In other words, an information producer will offer its initial product in different versions and leave the customers to select the version best suited to their needs. This strategy appeals to customers as well, because the customer can still have access to an information product even if he or she cannot afford the premium version. It is also a way for an information producer to capture a large share of the market. In order to execute this strategy, information firms simply create a high-end version of their product and create slightly lower-quality versions derived from it. Information producers like this strategy for two primary reasons. First, the low-end version can be used to advertise the qualitatively superior product. Many users of stripped-down encyclopedias, databases, or music-editing programs can readily see the value of buying an upgraded version of the software. For example, users of Adobe's free Reader will quickly come to realize the usefulness of being able to create PDF files or transfer PDF files into other file formats, which requires the priced version of Adobe Acrobat. Second, by offering lower-priced versions of the information product, a firm can build a reliable customer base.

Various business models have been tried in the information marketplace. We are also seeing new experimentation in the information marketplace with the advent of the Creative Commons and other open licenses. These open licenses have paved the way for Open Educational Resources

(OERs) in the education industry. These information goods are often produced and distributed by institutions, foundations, and governments to offset the price of education. Other industries have begun experimenting with open licensing and business structures as well, but, at least for now, education has mostly put this education model to work.

Currently, the most popular business models are some kind of subscription or paid service or generating revenues through advertising. Digital advertising, of course, depends on the targeted market segment an information provider is trying to appeal to. The most popular form of online marketing is known as personalized online marketing. This form of marketing is used by social media sites and other information providers such as Amazon, YouTube, and Google. Basic personal information is gleaned from the user's accounts and collected from the user's digital traces by storing and accessing the URLs of all the websites the user has visited. In the social media industry, companies like Facebook will track all the communication of a user, including very personal data, and post advertisements believed to be very appealing to the user. If the user is provided the option of clicking away an advertisement the social media site will simply use that data as an opportunity to place more personalized advertisements on the user's account. Personalized advertising has become the main source of monetization for social media platforms. However, the legitimate need for privacy protection is quickly at odds with this form of marketing. This unit explores these themes and the issues involved in our digital economy.

UNIT III

Digital Currency: Bitcoin and Blockchain

THIS UNIT FOCUSES ON THE RELATIVELY recent developments in digital economics called Bitcoin and blockchain. Sometimes, the terms *Bitcoin* and *blockchain* are used synonymously, but they are actually different tools (blockchain is now used for more than tracking digital cryptocurrencies). So first, we will examine what Bitcoin is. Bitcoin is a cryptocurrency that allows for the peer-to-peer transfer of money without the mediation of a bank or other financial institution. Cryptocurrency is simply a form of digital money that can be exchanged and transferred digitally on the Internet. For security reasons, the digital currency is encrypted. Bitcoin was invented by an anonymous individual or group of people called Satoshi Nakamoto. In 2008, Nakamoto published a paper titled “Bitcoin: A Peer-to-Peer Electronic Cash System,” in which the foundational principles of Bitcoin are developed and explained. In 2009, Nakamoto released Bitcoin as an open source code, and it was distributed across the world. Proponents of Bitcoin do not consider the actual identity of Nakamoto an important element of the Bitcoin system. Instead, they focus on the attributes of the Bitcoin currency, such as the fact that it is completely open and transparent, yet very secure, and that there is no central server, storage space, or government that can claim central authority over it.

Bitcoin has several interesting features. It allows users to bypass traditional banking methods and the fiat currencies used by nations. This way, users can avoid the fees and penalties imposed by most banks. The efficiency, speed, and security of Bitcoin transactions make Bitcoin an enticing option in the market of currencies. Furthermore, transactions using Bitcoin are completely anonymous, making it appealing to those concerned about personal privacy because there is no bank recording personal information. Funds are not tied to real-world financial institutions but rather digitally corresponding Bitcoin addresses. Owners of Bitcoin addresses are not explicitly identified, but all transactions on the blockchain are public. These benefits are made possible because it is by nature a decentralized open source project.

The technology that underlies Bitcoin is the public ledger known as blockchain. Transactions are recorded in the form of payer to payee using readily available software applications. Network computers can validate transactions and then add them to their copy of the ledger, and then broadcast these ledger additions to others in the network. To achieve independent verification of the chain of ownership each network node stores its own copy of the blockchain. Once a transaction is entered into the blockchain it is immune from revision. This is an important security feature that keeps others from tampering with the transactions and prevents what is known as the double-spending problem (the possibility of spending a digital currency twice). Each record has its own identifier and cannot be changed. This also builds a coherent and internal system of trust for those who use Bitcoin.

These are the themes explored in this unit. These readings serve as an entry-level guide to Bitcoin and blockchain. It is important to remember that Bitcoin has already demonstrated all the benefits and problems inherent to an open source project. It is true that many open source programs are very versatile, agile, and efficient. However, Bitcoin has undergone several forks, a phenomenon that often occurs when a group or community of developers disagree about the mission of the software (or foundation that has authority over the software) and takes the project in different directions. This can be very confusing for Bitcoin users and shows instability to the public at large. In a larger economic perspective, investing in Bitcoin can be seen as a high-risk investment; that is, an investment in Bitcoin is not like an investment in a real firm or product that provides tangible value in the marketplace. Furthermore, while Bitcoin has succeeded to build trust coherently within its system, many distrust it as a currency in the broader economy. Currencies are rarely used if they are not publicly trusted. At this point in history Bitcoin has developed interesting characteristics using innovative technology and it has solved some problems. However, it is unclear given its volatility whether it will gain larger public acceptance and not simply remain as a niche player in the digital marketplace.